Perturbing a bivariate-bicycle code cannot buy distance for free

An exact, perturbation-independent rate–distance obstruction for perturbed bivariate-bicycle (PBB) codes, with machine-checked certificates

Abstract

Perturbed bivariate-bicycle (PBB) codes are non-CSS stabilizer codes obtained from a CSS bivariate-bicycle (BB) parent P by adding a Z-type perturbation $[C\ D]$ to the X-check block. They have been proposed as a route past the CSS BB rate–distance envelope, on the strength of a published catalogue of 368 codes including a non-CSS $[\![144, 12, 12]\!]$ and a $[\![360, 12, \leq 24]\!]$ candidate.

We prove that the perturbation degree of freedom is not free. Let $\Delta = \{\lambda[C\ D] : \lambda[A; B] = 0\}$ be the *dressing space* of the perturbation. We show (i) Δ is an R-submodule of the Z-sector for $R = \mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$, (ii) the PBB dimension obeys the exact identity $k_Q = k_P - \dim \bar{\Delta}$, and (iii) any minimum-weight Z-logical of the parent that is *not* absorbed by Δ survives as a logical of Q, certifying $d_Q \leq d_Z(P)$.

Because Δ is a submodule, absorbing one minimum-weight parent logical absorbs its entire translation orbit. This upgrades the per-perturbation criterion to a **parent-level no-go theorem**: with $T(P) = \dim \big(M(P) + S_Z\big)/S_Z$, where $M(P)$ is the span of the translation orbits of *all* minimum-weight Z-logicals of P,

$$\boxed{d_Q > d_Z(P) \implies k_Q \leq k_P - T(P) \quad \text{for } \textit{every} \text{ perturbation } [C\ D]}$$

$T(P)$ depends only on the parent and is computed *exactly* — not bounded — by a SAT enumeration that terminates in a certified UNSAT. When $T(P) = k_P$ the entire family is closed: no PBB over P retains a single logical qubit while exceeding $d_Z(P)$.

We compute T exactly for 134 of the 202 distinct parents in the published catalogue — exhaustively for every parent at $n \leq 144$. 61 **parents are family-closed**, including the Gross code $A = x^3 + y + y^2,\ B = y^3 + x + x^2$ itself ($T = 12 = k_P$) and all 11 distinct catalogue $[\![144, 12, 12]\!]$ parents. This settles 249 of the 368 catalogue rows with no per-row search at all, and it forecloses the headline direction of the construction over the most important parent in the family.

The bound is not merely valid but *exact*. A second elementary identity — the dressing space is the image of the left kernel of $[A\ B]$, whose dimension is $k_P/2$ on every BB parent — caps $\dim \bar{\Delta} \leq k_P/2$. Whenever $T(P) \geq k_P/2$, which holds on 133 of the 134 certified parents, the two bounds **sandwich**:

$$\boxed{d_Q > d_Z(P) \text{ and } T(P) \geq k_P/2 \quad \implies \quad k_Q = k_P - T(P) \text{ exactly}}$$

Perturbation does not just pay at least T logical qubits for distance; on every parent in the theorem's class it pays *precisely* T, no more and no less. All 7 independently certified distance-increasing perturbations sit on the law exactly (slack 0, hypotheses replay-certified), and a controlled small-lattice probe — 64 sampled parents across four lattices, 26,898 valid $\delta > 0$ perturbations of Z-support ≤ 4, distances decided by exact symplectic meet-in-the-middle — found 496 further strict increases, **every one on the law, zero above it**: instances we now know were theorem-forced rather than lucky.

1 Setting and conventions

Let $R = \mathbb{F}_2[x,y]/(x^\ell - 1,\, y^m - 1)$ and let $n = 2\ell m$. A polynomial $f \in R$ acts as an $\ell m \times \ell m$ binary matrix via the commuting cyclic shifts x, y. For $A, B \in R$ the CSS **bivariate-bicycle** parent is

$$H_X = [\,A \;\; B\,], \qquad H_Z = [\,B^\mathsf{T} \;\; A^\mathsf{T}\,],$$

on n qubits split into two blocks of ℓm. Commutation is $H_X H_Z^\mathsf{T} = AB + BA = 0$, automatic since R is commutative.

A **perturbed bivariate-bicycle** (PBB) code adds a Z-type perturbation $[C\ D]$, $C, D \in R$, to the X-check block:

$$H_Q = \begin{pmatrix} A & B & | & C & D \\ 0 & 0 & | & B^\mathsf{T} & A^\mathsf{T} \end{pmatrix},$$

in the binary symplectic representation $(x \,|\, z)$ with symplectic product $\langle (x|z), (x'|z') \rangle_s = xz'^\mathsf{T} + zx'^\mathsf{T}$. Validity requires $M + M^\mathsf{T} = 0$ with $M = AC^\mathsf{T} + BD^\mathsf{T}$, i.e. **symmetry of** M, not its vanishing. (This is stated as $AC^\mathsf{T} + BD^\mathsf{T} = 0$ in some summaries; the weaker symmetric condition is the correct one and is what we verify.)

Throughout, $S_Z = \mathrm{rowspace}(H_Z) \subseteq \mathbb{F}_2^n$ is the parent's pure-Z stabilizer space, $k_P = n - \operatorname{rank} H_X - \operatorname{rank} H_Z$ is the parent dimension, k_Q the PBB dimension, and

$$d_Z(P) = \min\{\, \mathrm{wt}(z) : z \in \ker[A\ B] \setminus S_Z \,\}$$

is the parent's Z-distance. We write d_Q for the (symplectic) distance of Q and reserve *code distance* strictly for the algebraic quantity — never a decoder estimate, never a circuit or detector-graph distance.

Terminology. $C = D = 0$ recovers the CSS parent. We call a catalogue row a *reversal* when $d_Q > d_Z(P)$ is certified, and say the perturbation is *dominated* when $d_Q \le d_Z(P)$ is certified.

2 The dressing space and the dimension identity

Define the **dressing space**

$$\Delta \;=\; \big\{\, \lambda[C\ D] \;:\; \lambda \in \mathbb{F}_2^{\ell m},\ \lambda[A; B] = 0 \,\big\} \;\subseteq\; \mathbb{F}_2^n,$$

the set of Z-parts of those first-block row combinations whose X-part cancels, and write $\bar{\Delta} = (\Delta + S_Z)/S_Z$.

Theorem G. *For every parent P and every valid perturbation $[C\ D]$:*

(i) (Centralizer invariance.) *The pure-Z centralizer of Q equals that of P, namely* $\ker[A\ B]$.

(ii) (Dimension identity.) *The pure-Z stabilizer group of Q is $S_Z + \Delta$, and*

$$k_Q \;=\; k_P - \dim \bar{\Delta}.$$

(iii) (Survival criterion.) *If some minimum-weight $z \in \ker[A\ B] \setminus S_Z$ satisfies $z \notin S_Z + \Delta$, then $(0|z)$ is a nontrivial logical of Q of weight $d_Z(P)$, hence $d_Q \le d_Z(P)$.*

Proof. (i) A vector $(0|z)$ commutes with every second-block row automatically (Z against Z). Against a first-block row $(a\,|\,c)$ the symplectic product is $a \cdot z$. Hence $(0|z)$ is in the centralizer iff $[A\ B]z = 0$ — no dependence on $[C\ D]$.

(ii) A combination λ of first-block rows is pure Z iff its X-part $\lambda[A; B]$ vanishes, in which case it contributes exactly its Z-part $\lambda[C\ D]$. Adding the second block gives pure-Z stabilizer space $S_Z + \Delta$. For the dimension, project the row space onto the X-coordinate: the image is rowspace$[A\ B]$ for both codes, while the kernel is $\{0\} \times S_Z$ for H_P and $\{0\} \times (S_Z + \Delta)$ for H_Q — the pure-Z stabilizer space, by the first part of (ii). Hence $\operatorname{rank} H_Q = \operatorname{rank} H_P + \dim \bar{\Delta}$, and $k = n - \operatorname{rank} H$ gives $k_Q = k_P - \dim \bar{\Delta}$. (Note that rowspace H_P is *not* contained in rowspace H_Q — parent X-stabilizers are demoted — so the rank identity, not a containment, is what makes the count work.) Equivalently, via the sector identity valid for every stabilizer code, $\dim(\mathrm{Xcen}/S_X) = \dim(\mathrm{Zcen}/S_Z) = k$: the unchanged pure-$Z$ centralizer (i) minus the enlarged stabilizers $S_Z + \Delta$ is directly $k_P - \dim \bar{\Delta}$. (The shortcut "the X-sector of the quotient is unchanged, so only the pure-Z quotient loses dimension" is *false as an argument*: the pure-X centralizer also shrinks, $\mathrm{Xcen}(Q) = \mathrm{Xcen}(P) \cap \ker[C\ D]$ — e.g. from dimension 40 to 12 on `phase2_58` — and both the rank identity and the sector identity are what make the count work; `notes/theorem_j_xsector.md`, J.1–J.3.)

(iii) Immediate from (i) and (ii): $z \in \ker[A\ B]$ by (i), and $z \notin S_Z + \Delta$ means its class is nontrivial in the quotient of (ii). ☐

Part (iii) is the operational content: **a single unabsorbed minimum-weight parent logical caps the PBB distance**, with no search over PBB codewords. Machine checks of (i)–(iii) on catalogue instances are in `tests/test_pbb_survival.py`; the dimension identity (ii) holds exactly on all tested catalogue rows (§8).

3 From per-perturbation to parent-level: the module structure

The step that makes a family-wide theorem possible is the following.

Lemma 1 (submodule)**.** *Δ is an R-submodule of $\mathbb{F}_2^n$; so is S_Z, and hence so is $S_Z + \Delta$.*

Proof. The coefficient set $L = \{\lambda : \lambda[A; B] = 0\}$ is an ideal of R: if $\lambda A = \lambda B = 0$ then $(x^a y^b \lambda)A = x^a y^b(\lambda A) = 0$ and likewise for B, using commutativity of R. Since $\Delta = L \cdot [C\ D]$ and multiplication by $[C\ D]$ is R-linear, Δ is closed under the translations $x^a y^b$, i.e. is a submodule. $S_Z = \text{rowspace}[B^\mathsf{T}\ A^\mathsf{T}]$ is a submodule for the same reason. ☐

Corollary 2 (orbit absorption)**.** *If Δ absorbs a minimum-weight Z-logical z (i.e. $z \in S_Z + \Delta$) then it absorbs the entire translation orbit $\{x^a y^b z\}$ of z. Translations preserve weight and map $\ker[A\ B]$ to itself, so every orbit element is again a minimum-weight Z-logical.*

Now define, **from the parent alone**,

$$M(P) = \operatorname{span}\Big(\bigcup_z \operatorname{orbit}(z)\ :\ z \text{ a minimum-weight } Z\text{-logical of } P\Big), \quad T(P) = \dim \frac{M(P) + S_Z}{S_Z}.$$

Theorem H (parent-level no-go)**.** *For every parent P and every valid perturbation $[C\ D]$,*

$$d_Q > d_Z(P) \quad \Longrightarrow \quad k_Q\ \le\ k_P - T(P).$$

In particular, if $T(P) = k_P$ then no perturbation of P retains a logical qubit while exceeding $d_Z(P)$: the whole family over P is closed.

Proof. Suppose $d_Q > d_Z(P)$. By Theorem G(iii), contrapositively, **every** minimum-weight Z-logical z of P must be absorbed: $z \in S_Z + \Delta$. By Corollary 2 the whole orbit of each such z is absorbed, so $M(P) \subseteq S_Z + \Delta$, whence $\bar{M} \subseteq \bar{\Delta}$ and $\dim \bar{\Delta} \geq T(P)$. Theorem G(ii) gives $k_Q = k_P - \dim \bar{\Delta} \leq k_P - T(P)$. The final sentence is the case $T = k_P$, forcing $k_Q \leq 0$. □

Three features are worth emphasising. First, the hypothesis is *only* a distance increase — no assumption on C, D, on weight, or on locality. Second, $T(P)$ is a property of the parent, so one computation covers the infinite family of perturbations over P, including every $[C\ D]$ nobody has enumerated. Third, the conclusion bounds k_Q, not d_Q: the theorem does not forbid distance increase, it *prices* it.

3.1 Computing T exactly

T is obtained by a monotone SAT enumeration rather than estimated. Maintain $W = S_Z + M$, initially S_Z, and repeatedly ask a solver for $z \in \ker[A\ \ B]$ with $\mathrm{wt}(z) \leq d_Z(P)$ and $z \notin W$. Nontriviality with respect to W is encoded exactly as ordinary logical nontriviality — pair z against a basis of $W^{\perp}$, since $\exists f \in W^{\perp} : f \cdot z = 1$ iff $z \notin W$. Each SAT hit is necessarily a *minimum-weight* logical (nothing outside S_Z has weight below d_Z), and its full orbit is adjoined to M. The enumeration terminates in UNSAT, which certifies that W contains every minimum-weight Z-logical; therefore the resulting T is **exact**, not a lower bound. If the budget is exhausted first, the partial T is a certified *lower* bound — the sound direction, since Theorem H only needs $\dim \bar{\Delta} \geq T$.

Two soundness details are enforced in code. (a) Every SAT witness is re-verified through two independent GF(2) paths and its weight is checked to equal the supplied $d_Z(P)$ exactly; a mismatch is raised, never absorbed. (b) A translation symmetry break (anchor support at index 0 of one block) is applied to the decision queries. This is satisfiability-preserving precisely because $\ker[A\ B]$, S_Z and every M built from full orbits are translation invariant — the solution set of each query is translation invariant, so any solution can be translated to the anchored form. The premise is machine-validated in `tests/test_translation_symmetry_break.py` (automorphism action, transitivity, and 30 differential SAT/UNSAT agreements); it buys 6.2× on the decisive UNSAT proofs (77.8 s → 12.5 s on the $[\![144, 12, 12]\!]$ cap-11 instance).

3.2 The trade is exact: forced saturation

Theorem H bounds k_Q from above. A second, independent observation bounds $\dim \bar{\Delta}$ from above — and for most parents the two meet.

Lemma 2 (left-kernel identity)**.** *For every CSS BB parent, with* $L = \{\lambda : \lambda A = \lambda B = 0\}$ *the left kernel of* $[A\ B]$*,* $k_P = 2 \dim L$*.*

Proof. $\operatorname{rank} H_X = \ell m - \dim L$ (L is the left nullspace of H_X). The left nullspace of $H_Z = [B^{\mathsf{T}}\ A^{\mathsf{T}}]$ is the transpose of $K = \ker A \cap \ker B$, so $\operatorname{rank} H_Z = \ell m - \dim K$, and $k_P = \dim L + \dim K$. The coordinate-reversal map conjugates each cyclic shift to its transpose, so $\dim L = \dim K$. □

(Verified on all 202 *distinct parents,* `tests/test_pbb_theorems.py`*.)*

Lemma 3 (dressing ceiling)**.** $\Delta = L \cdot [C\ D]$ *is the image of a linear map on* L*, so* $\dim \bar{\Delta} \leq \dim L = k_P/2$ *for every valid perturbation.*

(Verified on all 368 *catalogue rows via* $\dim \bar{\Delta} = k_P - k_Q$*.)*

Theorem I (forced saturation)**.** *If* $T(P) \geq k_P/2$ *and* $d_Q > d_Z(P)$*, then* $\dim \bar{\Delta} = T(P)$ *and* $k_Q = k_P - T(P)$ ***exactly****.*

Proof. Theorem H gives $\dim \bar{\Delta} \geq T$; Lemma 3 gives $\dim \bar{\Delta} \leq k_P/2 \leq T$. Squeeze, then apply Theorem G(ii). □

Scope. $T \geq k_P/2$ holds on 133 of 134 certified parents. The single exception is `9a7638586033` ($n = 144$, $k_P = 12$, $T = 4$, not family-closed), where increase is only sandwiched to $4 \leq \dim \bar{\Delta} \leq 6$. On every other certified parent the exact trade law is a theorem: a distance increase pays precisely $T(P)$ logical qubits. This subsumes the empirical saturation of §5.2 — all 7 reversals have $T = k_P/2$ — and, retroactively, explains the small-lattice probe of §7: its 64 parents all have $T \in \{k_P/2, k_P\}$, so its 496 strict increases could not have landed anywhere but on the law.

4 The Gross code: its entire perturbation family is closed

The canonical Gross code (Bravyi et al. [1]) is the BB code with $\ell = 12$, $m = 6$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, giving $[\![144, 12, 12]\!]$. Running the enumeration of §3.1 on it:

Table 1: The Gross-code parent.

parent	catalogue row	ℓ, m	A	B	k_P	$d_Z(P)$	T	T exact	family closed
`aebb649586c1f5ab`	`12_6_0194`	$12, 6$	$x^3 + y + y^2$	$y^3 + x + x^2$	12	12	**12**	yes (UNSAT)	**yes**

$T = k_P = 12$, so by Theorem H:

Corollary 3 (Gross closure)**.** *No perturbed bivariate-bicycle code built on the Gross code retains a single logical qubit while exceeding Z-distance* 12. *The entire perturbation family over the Gross code — not merely the catalogued members — is closed.*

The same holds for **all** 11 **distinct** $[\![144, 12, 12]\!]$ **parents present in the published catalogue**, each with $T = 12 = k_P$ certified by a terminating UNSAT, including `4c7eb964a6a3e38c`, the parent of the benchmark non-CSS PBB $[\![144, 12, 12]\!]$ (`12_6_0193`, 220.4 s, 3 witnesses). That benchmark code is independently interesting — we certify it is *genuinely* non-CSS (not CSS after row operations, any qubit permutation, or any local Hadamard; a 19-parity-mask plus one-hot certificate is CP-SAT infeasible) and that its exact distance is $d = 12$, two-sided: weights ≤ 5 excluded by exact meet-in-the-middle, weight 6 by a complete classification (all 72/72 zero-syndrome vectors coincide with the stored pure-Z checks, so no weight-6 logical exists), weights 7–11 by four CaDiCaL UNSAT sector proofs (502/674/422/576 s), and a weight-12 witness verified through two independent GF(2) paths. It matches the Gross code's $[\![144, 12, 12]\!]$ parameters; Corollary 3 shows that no sibling perturbation over the same parent can improve on them without giving up all 12 logical qubits.

5 Catalogue-scale results

The published catalogue [2] contains 368 PBB rows over 202 distinct CSS BB parents. Applying §3.1 to the parents:

The per-length breakdown matters more than the totals, because the enumeration is **complete at every length up to** $n = 144$:

So over the entire catalogue at $n \leq 144$ — which contains every $[\![144, 12, 12]\!]$ candidate and both $[\![72, \cdot, \cdot]\!]$ reversals — the classification is exhaustive: all 117 parents have an exact T, 57 of them are family-closed, and 224 of 254 rows are settled with no per-row search. The 68 uncertified parents are exactly the $n = 180$ and $n = 360$ ones, where a single terminating UNSAT can cost hours (the

quantity	value
distinct parents	202
parents with **exact** T (terminating UNSAT)	134
parents **family-closed** ($T = k_P$)	**61**
catalogue rows capped *a priori* ($k_Q > k_P - T$, no per-row work)	**249** of 368
catalogue rows Theorem H permits, of the 279 classified (candidates for a genuine increase)	30

n	parents	certified	family-closed	rows	capped *a priori*
36	3	3/3	3	47	47
72	8	8/8	5	34	32
108	68	68/68	28	107	80
144	38	38/38	21	66	65
≤ 144	117	117/117	57	254	224
180	46	16/46	4	64	24
360	39	1/39	0	50	1

slowest completed parent took 16,493 s); that sweep continues, and because T is monotone under further enumeration these figures are lower bounds on the final closure count, never upper bounds.

5.1 A structural confinement

Cross-tabulating the 134 certified parents against their 279 catalogue rows exposes a sharp pattern. Only four ratios k_Q/k_P occur, and only three ratios T/k_P:

k_Q/k_P	$T/k_P = 1$	$T/k_P = 1/2$	$T/k_P = 1/3$	capped *a priori*
1	109	67	1	yes
5/6	11	10	—	yes
3/4	25	13	—	yes
1/2	13	**30**	—	13 yes / **30 no**

Every row that keeps more than half of the parent's logical qubits is capped *a priori*: $d_Q \leq d_Z(P)$ with no search. The only rows Theorem H permits to increase distance are the k-halving ones with $T/k_P = 1/2$ — exactly 30 of 279. The observed distance-increasing perturbations are confined to a k-halving class, and this is now a proved confinement rather than an empirical regularity.

5.2 The bound is saturated, not merely valid

The catalogue contains 7 perturbations for which we hold a certified reversal $d_Q > d_Z(P)$. Theorem H must permit every one of them or it is refuted. A refutation requires *both* halves — hypothesis and violated conclusion — so we verify the hypothesis independently by a replayed UNSAT lower bound on d_Q, and count any check whose hypothesis is not independently established as vacuous rather than as support.

7/7 checked, 7/7 non-vacuous, 0 violations, and **slack exactly** 0 **in every case**: $k_Q = k_P - T(P)$ identically. Every certified distance increase pays exactly the price Theorem H charges, never less and never more — and Theorem I (§3.2) proves they never could: all seven have $T = k_P/2$, inside the theorem's class, so equality was forced before it was observed.

row	n	k_P	k_Q	$d_Z(P)$	certified $d_Q >$	T	ceiling $k_P - T$	slack	hypothesis
`12_6_0217`	144	8	4	8	9	4	4	**0**	certified
`phase2_58`	72	8	4	4	4	4	4	**0**	certified
`phase2_60`	72	8	4	4	4	4	4	**0**	certified
`phase2_71`	108	12	6	4	5	6	6	**0**	certified
`phase2_72`	108	12	6	4	5	6	6	**0**	certified
`phase2_88`	108	4	2	4	5	2	2	**0**	certified
`9_6_0183`	108	4	2	8	9	2	2	**0**	certified

Observation 4 (saturation — now a theorem for $T \geq k_P/2$). On every certified reversal in the published catalogue, $k_Q = k_P - T(P)$. Theorem I (§3.2) proves this is forced for every parent with $T \geq k_P/2$ — all certified parents except `9a7638586033`; §7 carries the residual.

The gate is executable: `experiments/exp039_nogo_module.py gate` re-derives the table and exits nonzero on any violation, and is locked by `tests/test_pbb_nogo.py`.

6 What this does and does not settle

Settled. Over the 61 family-closed parents — including the Gross code and every catalogue $[\![144, 12, 12]\!]$ parent — the perturbation degree of freedom cannot produce a code that beats the parent's Z-distance while retaining a logical qubit. This is a theorem about all $[C\ D]$, not a search result, and it is certified by terminating UNSAT proofs with hash-bound, replayable CNFs.

Priced, not forbidden. For the remaining parents, distance increase is possible but pays at least $T(P)$ logical qubits, and — by Theorem I — pays *exactly* $T(P)$ on every parent with $T \geq k_P/2$, which is every certified parent but one. It would be wrong to claim the trade is always unprofitable *relative to the parent*: on five of the seven certified reversals it is mildly profitable in kd^2/n, because $+2$ distance at $d_Z(P) = 4$ outweighs halving k (e.g. `phase2_71`: parent $12{\cdot}4^2/108 = 1.78$ versus PBB $6{\cdot}6^2/108 = 2.00$). The correct comparison is not the parent but the CSS envelope at the same length, and there the reversals lose decisively — every one of the seven is dominated at equal n by a CSS BB code with $k_C \geq k_Q$ and certified exact $d_C \geq$ the PBB's upper bound (all seven machine-checked in `results/processed/exp036_envelope_check.json`, schema `exp036-envelope-check-v2`, against 162 certified-exact CSS candidates; the dominator shown is the strongest at that length by kd^2):

reversal	$[\![n, k_Q, d_Q{\leq}]\!]$	kd^2/n	dominating CSS	kd^2/n	factor
`12_6_0217`	$[\![144, 4, 10]\!]$	2.78	$[\![144, 12, 12]\!]$	12.00	4.3×
`9_6_0183`	$[\![108, 2, 10]\!]$	1.85	$[\![108, 8, 10]\!]$	7.41	4.0×
`phase2_71`, `phase2_72`	$[\![108, 6, 6]\!]$	2.00	$[\![108, 8, 10]\!]$	7.41	3.7×
`phase2_88`	$[\![108, 2, 6]\!]$	0.67	$[\![108, 8, 10]\!]$	7.41	11.1×
`phase2_58`, `phase2_60`	$[\![72, 4, 6]\!]$	2.00	$[\![72, 12, 6]\!]$	6.00	3.0×

So the perturbation can beat *its own parent*, but never the best CSS code of the same length: the reversals trade a 3.0–11.1× deficit against the envelope for a $\leq 1.13\times$ gain against the parent.

The X sector. The full analysis is in `notes/theorem_j_xsector.md`; what the main results need is: $d_X(P) = d_Z(P)$ for every BB parent (reversal symmetry, J.0), so the Z-side framing loses no generality; the pure-X centralizer of Q shrinks by $\rho_X = \dim((R_{CD} + S_Z)/S_Z) \geq \dim \bar{\Delta}$ (J.1); and Q's pure-X logicals are either survivors of weight $\geq d_X(P)$ or demoted parent stabilizers paired with parent Z-logicals (J.4). Whether $d_X(Q) \geq d_X(P)$ holds unconditionally is **open**: zero violations in 6.47M valid perturbations across 12 lattices (counts after a GF(2)-matmul correction of an earlier uint8 filter bug, `notes/theorem_j_xsector.md` §3); on the 221 catalogue rows where both $d_X(Q)$ (exact) and $d_X(P) = d_Z(P)$ (certified) are known, 17 at equality and 0 violations ($d_X(Q)$ exact on 276 of 368 rows); and on the 116 **parents at** $n \leq 144$ — every parent at $n \leq 144$ — EXP-044 checked 5,483 valid instances against certified $d_Z(P)$: 4,009 provably non-decreasing, 1,474 honestly capped-unresolved (kernel dimension > 18), and *zero* decrease witnesses, with $\min(w_{\text{dem}} - d_Z(P)) = 0$ attained but never negative — but no proof yet. The caps in Theorems G/H/I do not require it: they cap d_Q through pure-Z survivors only.

The residual class is small. The known violation of $T \geq k_P/2$ might still be the tip of a large uncharted class. It is not, at least at small lattice size: among 10,645 distinct connected $k_P \geq 2$ parents with A, B each of weight ≤ 3 (support families 2×2, 2×3, 3×3), exhaustive over all 43 ordered lattices with $\ell \cdot m \leq 40$ and quotiented by independent block translations with $\{A, B\}$ unordered (4,621 parents for $\ell{\cdot}m \leq 24$ plus 6,024 for $24 < \ell{\cdot}m \leq 40$; lattices 6×7 and larger skipped by an extrapolated budget model), *none* has $T < k_P/2$ — in every case computed, $T \in \{k_P/2,\, k_P\}$ exactly. The only known exception (`9a7638586033`, k_P=12, T=4) sits at $\ell \cdot m = 72$, outside this scope. Whether the residual class (parents with $T < k_P/2$) is empty beyond `9a7638586033` at larger lattices is under computation; Conjecture B′ (§7) governs what happens on any residual parent that does exist.

Not settled by this paper. (i) Whether *every* $\delta > 0$ PBB is dominated by *some* CSS code — the universal envelope statement — remains open; 60 catalogue rows are pending and 30 are flagged as candidate escapees — all pool-coverage artifacts (no certified same-length CSS code exists at that k in our pool yet) rather than verified escapes; the pending count only decreases, and the escapee count shrinks as the CSS pool grows — both are provisional. (ii) The $n = 360$ parents remain open (38 of 39 not yet certified). (iii) Everything here is a *code-level* statement. Circuit-level cost is a separate axis: our detector-error-model mechanism-distance intervals for the two $[\![144, 12, 12]\!]$ circuits are tied at the certified interval $[5, 12]$ on each — weight-4 exclusion completed on both circuits (perfect-matching class by meet-in-the-middle, star class by exact cover; no logical failure from any set of ≤ 4 DEM mechanisms; structural-probability DEM), with weight-12 data-logical witnesses as upper bounds (EXP-029/EXP-040) — the PBB schedule is $\approx$ 6–17× slower on an idle machine (p_{50} 10.3×, p_{95} 16.6×, p_{99} 6.3×), and no circuit-level result has favoured the PBB candidate. A verified one-ancilla mixed-stabilizer syndrome-extraction circuit for the non-CSS PBB, benchmarked against the Gross code under identical noise, remains the decisive end-to-end experiment.

Honest limitations. T is exact only where the enumeration terminated in UNSAT (134/202 parents; all 117 at $n \leq 144$); elsewhere it is a certified lower bound, which is the sound direction. $d_Z(P)$ is used as supplied and is independently certified for the parents in the table. Catalogue-reported distances are treated as untrusted throughout; where we quote a distance we quote our own two-sided certificate, and one catalogue heuristic bound was found to be $\geq 2.5\times$ loose ($d \leq 40$ advertised, ≤ 16 re-verified).

7 The exact trade law, its residual, and remaining conjectures

Theorem I (restated; proved in §3.2). *If $T(P) \geq k_P/2$ and $d_Q > d_Z(P)$, then $k_Q = k_P - T(P)$ exactly. This covers* 133 *of* 134 *certified parents and all* 57 *family-closed ones at $n \leq 144$; the saturation observed in §5.2 and the* 496/496 *small-lattice increases are instances of it.*

Three layers of machine evidence now sit beneath the theorem and delimit its residual:

1. *Catalogue upper law.* On all 279 classified rows, $\dim \bar{\Delta} \leq T(P)$ with zero exceptions (spread

of $\dim \bar{\Delta} - T$ only non-positive, down to -24). For $T \geq k_P/2$ parents this is Lemma 3; the law holding also on the one $T < k_P/2$ parent's rows is evidence the residual class is benign.

2. *Small-lattice probe (EXP-040).* 64 parents across four lattices, 26,898 unseen $\delta > 0$ perturbations: $\dim \bar{\Delta} < T$ on 23,634, $= T$ on 3,264, $> T$ on **zero**; all 496 strict increases at $= T$, with actual containment $\bar{M} = \bar{\Delta}$ verified, not just dimensions. All probed parents had $T \in \{k_P/2, k_P\}$, so these are theorem-forced.

3. *Reversals.* All 7 catalogue reversals at $\dim \bar{\Delta} = T = k_P/2$ with $\bar{M} = \bar{\Delta}$.

Conjecture B′ (residual saturation)**.** *For parents with $T < k_P/2$ (one known: `9a7638586033`, where increase sandwiches to $4 \leq \dim \bar{\Delta} \leq 6$), distance increase still forces* $\dim \bar{\Delta} = T$.

Evidence: the catalogue upper law above, and the probe's zero violations; *the proof must beat Lemma 3's ceiling, which is no longer tight there.*

Conjecture C (k-halving confinement)**.** *Any perturbation with $d_Q > d_Z(P)$ has $k_Q/k_P = 1/2$. On every $T = k_P/2$ parent this is immediate from Theorem I; the conjecture is residual exactly on $T < k_P/2$ parents and on the 68 uncertified ones ($n \in \{180, 360\}$). Every one of the 279 classified rows with $k_Q/k_P \in \{1, 3/4, 5/6\}$ is capped* a priori *(§5.1, zero exceptions).*

Conjecture D (distance parity)**.** *Certified reversals gain exactly $+2$ in distance.*

Evidence: the uniform gap $d_Q - d_Z(P) = 2$ across all 7 reversals, whose exact d_Q values are certified; consistent with d_Z even throughout. (The probe stores only the increase decision, not the achieved d_Q, so its 496 increases are not parity evidence.)

The open problem that remains is not *whether* perturbation pays — it pays exactly T — but *whether anything on the CSS side of the envelope can ever be beaten this way*: §6's domination table says no for all seven known increases.

8 Reproduction and provenance

Environment: macOS Darwin 25.5.0 arm64, Apple M3 Ultra, 28 logical CPUs, 96 GiB; Python 3.13.9 in `.venv`; `numpy 2.4.6`, `scipy 1.18.0`, `stim 1.16.0`, `pymatching 2.4.0`, `sinter 1.16.0`, `ldpc 2.4.1`, `galois 0.4.11`, `ortools 9.15.6755`, `python-sat 1.9.dev13`; solver CaDiCaL 1.9.5 via PySAT. Full suite: 898 passing tests, 1 skipped (899 collected).

Every SAT decision records its canonical CNF SHA-256 and encoding version; verdicts are rederived from rebuilt matrices on replay, and stamps that fail to hash-bind are rejected rather than trusted. Forgery regression tests confirm that crude tampering dies arithmetically or under replay.

Reproduce the headline result:

```
cd math/qec
PYTHONPATH=src .venv/bin/python experiments/exp039_nogo_module.py run --ns 144
PYTHONPATH=src .venv/bin/python experiments/exp039_nogo_module.py gate
PYTHONPATH=src .venv/bin/python -m pytest tests/test_pbb_nogo.py tests/test_pbb_survival.py -q
```

9 Related work and novelty

Bivariate-bicycle codes and the Gross code are due to Bravyi et al. [1]. The PBB construction and the 368-row catalogue are from [2]. Cruz-Benito et al. [2] (v1, Sec. III.2, Lemma 0) give

claim	artifact	experiment
Theorem G, machine-checked on catalogue rows	`src/qec_research/codes/pbb_survival.py`, `tests/test_pbb_survival.py` (15 checks)	EXP-038
Theorem H, module + exact T enumeration	`src/qec_research/codes/pbb_nogo.py`, `tests/test_pbb_nogo.py` (17 checks)	EXP-039
Gross closure, 61 closed parents, 249 rows capped, complete at $n \leq 144$	`results/partial_runs/exp039_nogo_module.json`	EXP-039
Saturation table, 7/7 non-vacuous, gate	`exp039_nogo_module.py` gate	EXP-039
Certified d_Q lower bounds for reversals	`results/partial_runs/exp037/row_*.json`	EXP-037
Theorem I legs: $k_P = 2\dim L$ (202 parents), $\dim\bar{\Delta} \leq k_P/2$ (368 rows), scope exception	`tests/test_pbb_theorems.py`, `results/processed/exp040_saturation_probe.json`	EXP-039/040
All 7 reversals CSS-dominated at equal n (162 exact candidates)	`results/processed/exp036_envelope_check.json`	EXP-036/037
5 certified reversals, 41 dominations of 87	`results/partial_runs/exp036_delta_closure.json`	EXP-036
Exact $d = 12$ for `12_6_0193`, two-sided	`results/certificates/pbb_12_6_0193_distance.json`	EXP-035
Genuinely non-CSS (`12_6_0193`)	`notes/novelty_matrix.md`, EXP-034 records	EXP-034
Catalogue-wide CSS-envelope classification	`results/partial_runs/exp037_envelope_classification.json`	EXP-037
Strict-provenance $\delta > 0$ audit (155 rows)	`results/processed/exp027_delta_audit.json`	EXP-027
Translation symmetry break soundness, 6.2×	`tests/test_translation_symmetry_break.py`	FR-022
DEM mechanism-distance intervals, LER, latency	`reports/technical_report.md` §5–6	EXP-016/029

the validity condition; we derive it independently from the binary symplectic Gram matrix and confirm the correct condition is *symmetry* of $AC^\mathsf{T} + BD^\mathsf{T}$, not its vanishing. Multivariate bicycle generalisations are in [4]; existence and characterisation results for BB codes in [3]. ASC [7] certifies no-depth-6 for IBM BB codes; our depth criterion is independent and basis-independent.

To our knowledge the following are new here: (a) the identification of Δ as an R-submodule and the resulting orbit-absorption argument; (b) the exact dimension identity $k_Q = k_P - \dim\bar{\Delta}$ as a certified structural fact rather than an observed coincidence; (c) the perturbation-independent no-go bound (Theorem H) and its exact T enumeration with a terminating UNSAT certificate; (d) the closure of the entire perturbation family over the Gross code; (e) the saturation observation $k_Q = k_P - T$ on all certified reversals; and (f) the proved confinement of distance increase to a k-halving class.

10 Conclusion

The PBB construction's extra freedom is real but not free. It is governed by a single submodule Δ whose dimension is exactly the dimension deficit, and whose module structure forces any distance increase to absorb entire translation orbits of the parent's minimum-weight logicals. This yields a computable, perturbation-independent obstruction that closes 61 of 202 published parents outright — the Gross code among them — caps 249 of 368 catalogue rows with no search, confines distance increase to a k-halving class, and is saturated with slack exactly zero on all 7 certified distance-increasing perturbations known to us.

The abstract $[\![n, k, d]\!]$ route past the CSS bivariate-bicycle envelope is, over these parents, closed by theorem. What remains genuinely open is the circuit-level question: whether a verified fault-tolerant syndrome-extraction circuit for a non-CSS PBB can beat the Gross code end-to-end under identical noise. Nothing in our circuit-level evidence so far favours it.

References

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[2] Juan Cruz-Benito, Andrew W. Cross, David Kremer, and Ismael Faro. Evolutionary discovery of bivariate bicycle codes with LLM-guided search, 2026. arXiv:2606.02418.

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[4] Lukas Voss, Sim Jian Xian, Tobias Haug, and Kishor Bharti. Multivariate bicycle codes, 2024. arXiv:2406.19151.

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[7] Kai Zhang, Dingchao Gao, Zhaohui Yang, Runshi Zhou, Fangming Liu, Zhengfeng Ji, and Jianxin Chen. Optimal compilation of syndrome extraction circuits for general quantum LDPC codes, 2026. arXiv:2603.21499.